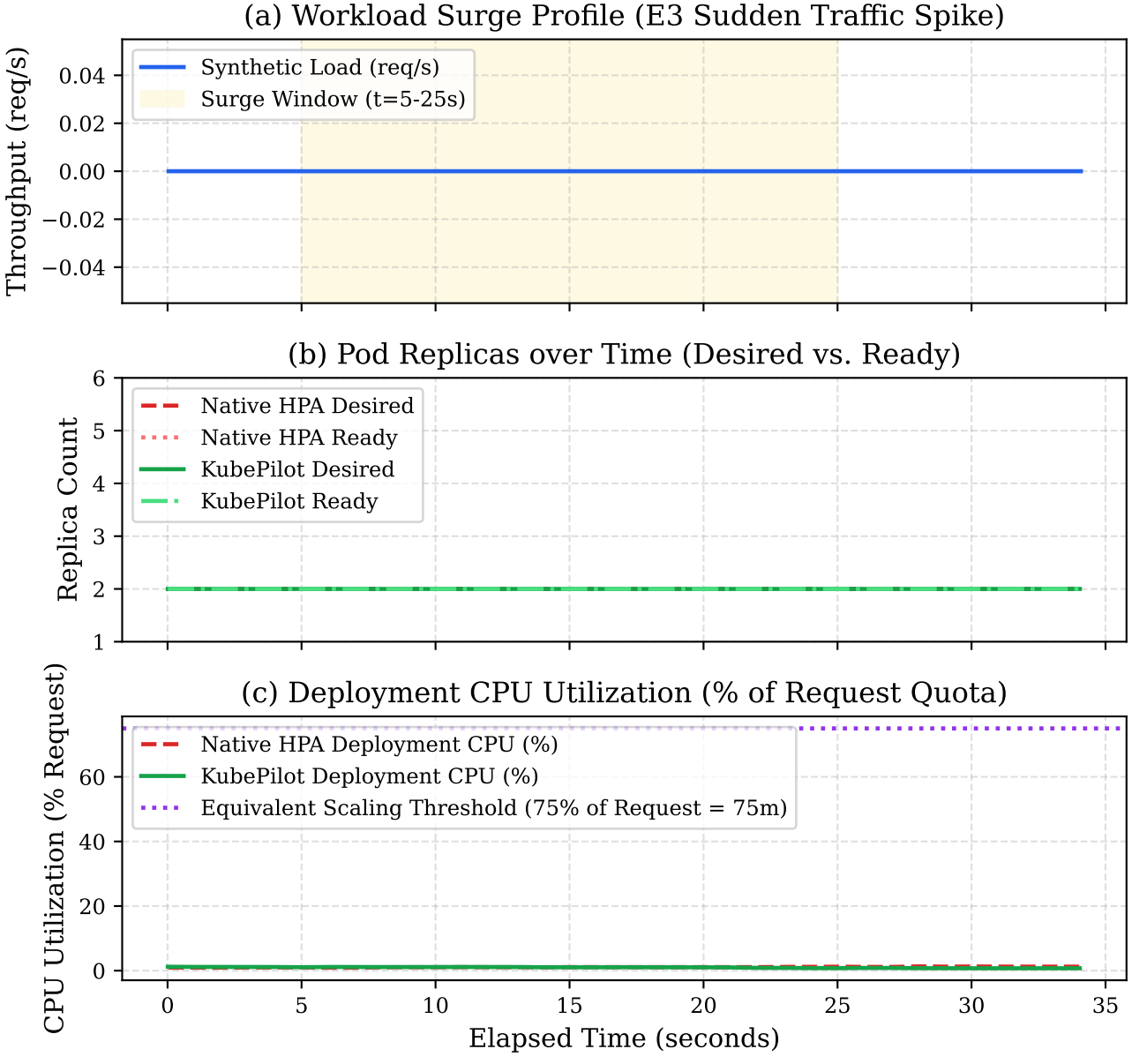
(a) Workload Surge Profile (E3 Sudden Traffic Spike)
Throughput (req/s)
Synthetic Load (req/s)
Surge Window (t=5-25s)
0.04
0.02
0.00
−0.02
−0.04
(b) Pod Replicas over Time (Desired vs. Ready)
Replica Count
Native HPA Desired
Native HPA Ready
KubePilot Desired
KubePilot Ready
6
5
4
3
2
1
(c) Deployment CPU Utilization (% of Request Quota)
CPU Utilization (% Request)
Native HPA Deployment CPU (%)
KubePilot Deployment CPU (%)
Equivalent Scaling Threshold (75% of Request = 75m)
60
40
20
0
0
5
10
15
20
25
30
35
Elapsed Time (seconds)